
Computergrafik und Computeranimation

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1. Einführung in die Computergrafik
 - 1.1 Computergrafik – Begriff, Vektorgrafik, Abgrenzung zur Rastergrafik
 - 1.2 Historie und Anwendungsfelder
 - 1.3 Mathematische Grundlagen



1.1. Einführung in die Computergrafik

- (I) Computergrafik – Begriff
- (II) Vektorgrafik vs. Rastergrafik
- (III) Rendering-Pipeline



(I) Computergrafik – Begriff

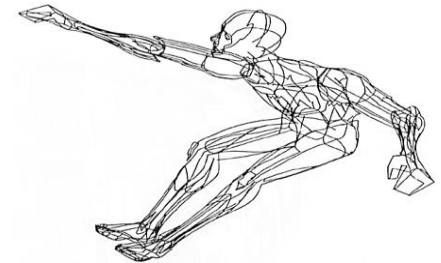
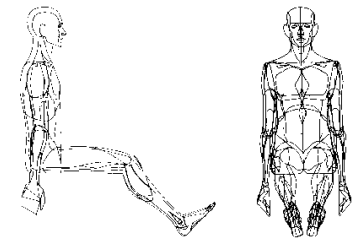
“Perhaps the best way to define *computer graphics* is to **find out what it is not**. It is not a machine. It is not a computer, nor a group of computer programs. It is not the know-how of a graphic designer, a programmer, a writer, a motion picture specialist, or a reproduction specialist.

Computer graphics is all these – a consciously managed and documented **technology directed toward communicating information accurately and descriptively.**”

Computer graphics, William A. Fetter, 1966



William A. Fetter



“Boeing Man”

(I) Computergrafik – Begriff

“*Computer graphics* (CG) is the field of visual computing, where one utilizes computers both to generate visual images synthetically and to integrate or alter visual and spatial information sampled from the real world.”

https://en.wikipedia.org/wiki/Category:Computer_graphics, Abruf 5.10.20

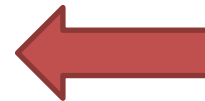
“*Computer graphics* is the branch of computer science that deals with generating images with the aid of computers.”

https://en.wikipedia.org/wiki/Computer_graphics, Abruf 5.10.20

1.1. Einführung in die Computergrafik

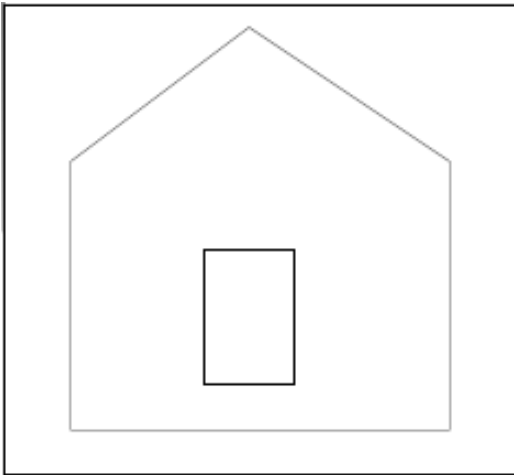
(I) Computergrafik – Begriff

(II) Vektorgrafik vs. Rastergrafik

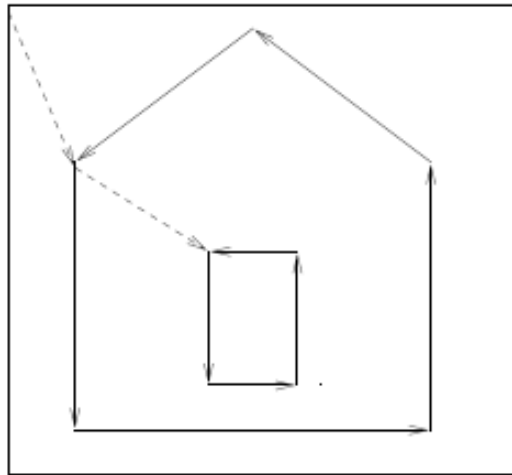


(III) Rendering-Pipeline

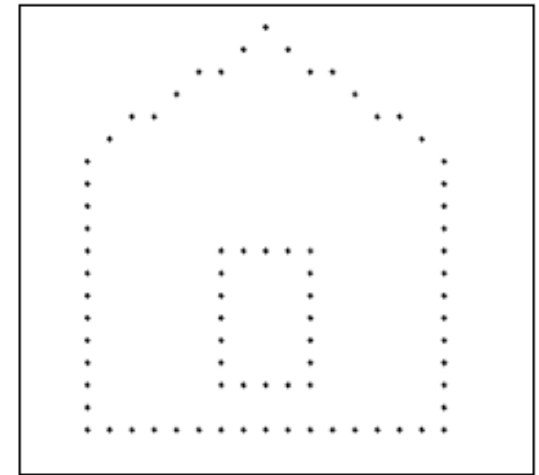
(II) Vektorgrafik vs. Rastergrafik



Originalbild



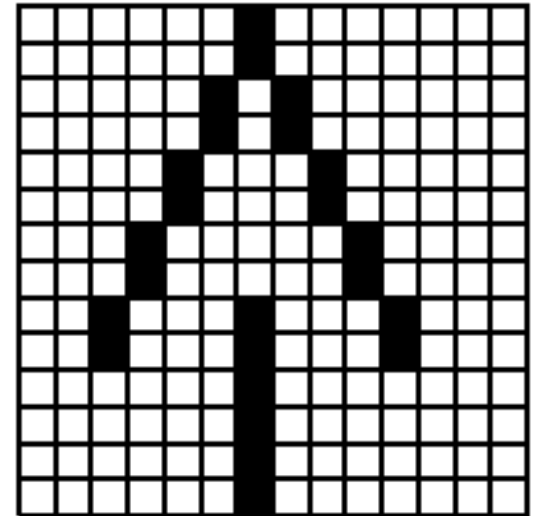
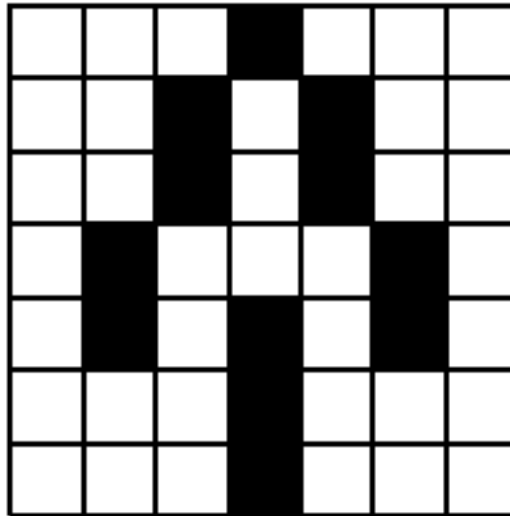
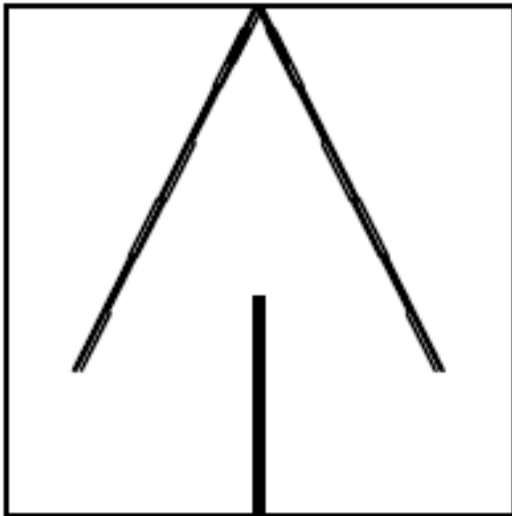
Vektorgrafik:
Geometrische
Grundobjekte



Rastergrafik:
Pixelmatrix

(II) Vorteile von Vektorgrafik

- Vorteil: Beliebig skalierbar ohne Qualitätsverlust
- Umrechnung zur Darstellung auf Rasterbildschirm (Scan Conversion, Rasterung)
- Problematisch: Aliasing

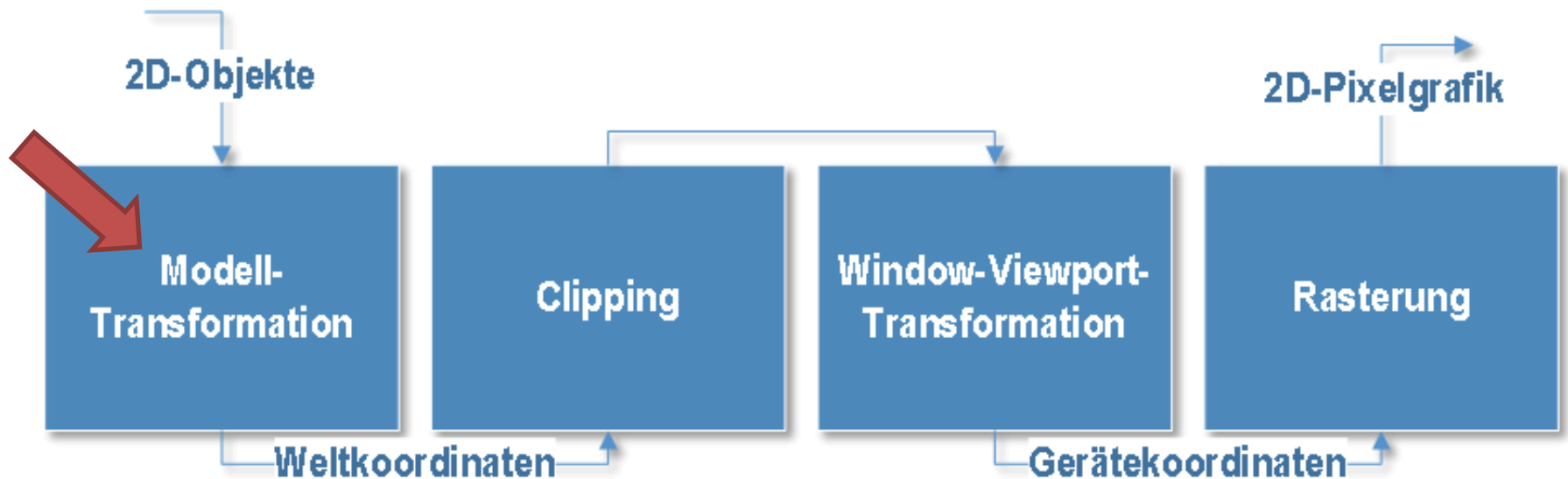


1.1. Einführung in die Computergrafik

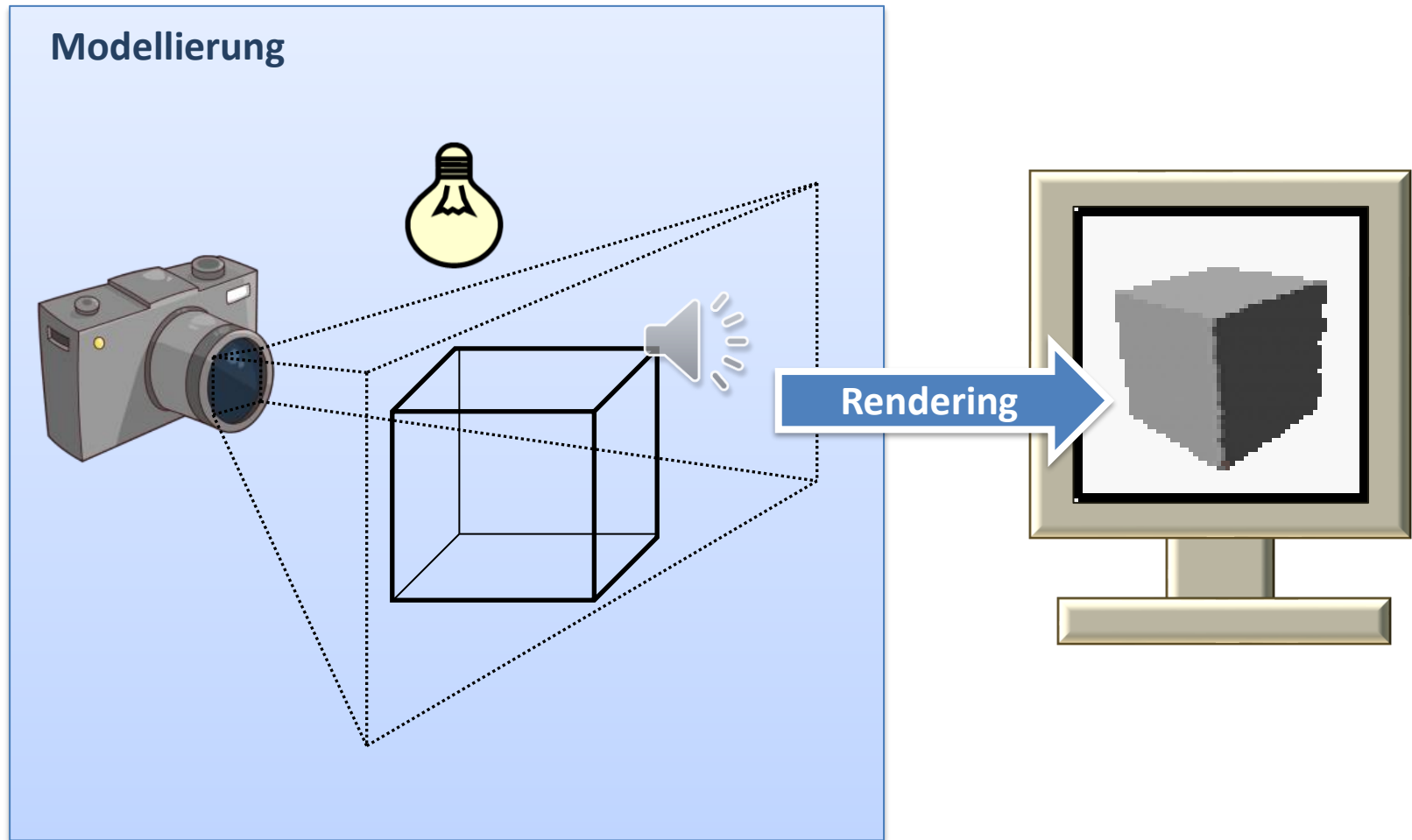
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(III) Vereinfachte 2D-Rendering-Pipeline



(III) 3D-Computergrafik



(III) Klassische 3D-Rendering-Pipeline (vereinfacht)

